

Akash Gogate

Princeton, NJ • 609-751-2597 • akash.gogate@gmail.com • agogate@wisc.edu • [linkedin.com/in/akashgogate](https://www.linkedin.com/in/akashgogate) • github.com/AkashGogate

EDUCATION

University of Wisconsin at Madison

Graduating: May 2027

Bachelor of Science in Computer Science • Dean's List Fall 2024, Spring 2025, Fall 2025 • 3.9 GPA

Relevant Coursework: Artificial Intelligence, Deep Learning, Advanced Algorithms, Machine Organization and Programming, Cryptography, Data Structures

South Brunswick High School

Relevant Coursework: Computer Science Capstone, Android Application Development, Virtual Reality, Game Development, Web Development

TECHNOLOGIES

Languages: Python, R, Java, JavaScript, Rust, C++, SQL, Linux/Bash

ML & AI: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, OpenCV, Mediapipe, Sentence-Transformers, Hugging Face, Transformer Architectures, NLP, Random Forest, KNN, Feature Engineering, Hyperparameter Tuning, Model Fine-Tuning, Inference Optimization, Model Deployment

Agentic & LLM: LLM Agents, LLM APIs, RAG, Semantic Similarity Search, Vector Databases (Supabase, Pinecone, Qdrant), Claude Code, Codex, AgentBench, terminal-bench, FastAPI, MLflow, Weights & Biases, A/B Testing, Benchmarking

GPU & MLOps: CUDA, GPU-Accelerated Computing, Docker, Kubernetes, Kafka, MongoDB, CI/CD, GitHub Actions, Git, Linux, pytest, Experiment Tracking, Model Monitoring, Model Serving, Inference Serving

EXPERIENCE

The Kendziorz Lab at the University of Wisconsin-Madison

September 2025 - Present

Machine Learning Research Intern

- Built end-to-end ML pipeline for spatial transcriptomics and scRNA-seq (NGS) gene expression analysis; agentic framework selects and optimizes models across scVI, scANVI, scGen, and AmortizedLDA, **reducing GPU computation by 300%**; tracked experiments and model performance using Weights & Biases
- Engineered TransferAgent module automating citation mapping, peer-reviewed literature synthesis, and cross-paper validity scoring for an agentic academic grading framework, cutting review time by **~60%**; collaborated with clinicians on clinical genomics impact assessment

Leidos

May 2025 - Present

Software Engineer Intern

- Optimized satellite scheduling via memoization + DP in Python, **reducing constraint violations**; designed low-latency Kafka + MongoDB event pipeline with **fault-tolerant streaming** and at-least-once delivery guarantees for mission-critical ops
- Deployed containerized test suite on Kubernetes via **GitHub Actions CI/CD** with comprehensive unit coverage across all system components

Inspirit AI Research Internship

Sep 2022 - Mar 2023

Machine Learning Research Intern

- Built **skin cancer detection pipeline** using OpenCV preprocessing and feature extraction fed into scikit-learn classifiers (Decision Tree, KNN) achieving **93% classification accuracy** via hyperparameter tuning and cross-validation; published as **peer-reviewed paper** through Inspirit AI
- Separately trained a Random Forest classifier on miRNA sequences achieving **95% predictive accuracy ($p < 0.05$)** for cancer cell likelihood prediction, validated via stratified k-fold cross-validation across multiple data splits

TECHNICAL PROJECTS

Self-Improving LLM Agent (<https://github.com/AkashGogate/SelfImprovingLLMAgent>)

Jan 2026 - May 2026

- Architected failure-driven self-improvement loop for LLM agents across **5,000+ task benchmark**, outperforming ReAct baseline by **15pp** via RAG-enhanced retrieval, Transformer-based reasoning, and failure-case feedback loops
- Optimized model inference via task horizon length tuning and inference parameter search; built provider-agnostic model serving abstraction layer supporting end-to-end ML pipeline from data ingestion to inference, scaled to task horizon 20 with GitHub Actions CI/CD
- Engineered custom evaluation infrastructure: AgentBench, terminal-bench, and 6-task deterministic suite with LLM-as-judge harness scoring 7 dimensions and matplotlib visualization pipeline; implemented novel repeated_failure_rate metric to quantify self-improvement efficacy

Lotus Health (<https://github.com/AkashGogate/lotus-health>)

March 2026

- Financial risk modeling engine on **1,080-node ICD-10 comorbidity graph** integrating MEPS and CMS insurance data to compute 5-year disease cost paths across **45M records**
- Implemented RAG pipeline using semantic similarity search over ICD-10 graph (SQL-queried NoSQL store) as vector database; object-oriented Python NLP layer separates inference from deterministic logic, eliminating hallucinations in outputs

miRcore Cancer Detection Research

Jul 2021

- Trained Random Forest classifier in R on micro-RNA expression sequences achieving **95% predictive accuracy ($p < 0.05$)** for cancer cell likelihood prediction, validated via stratified k-fold cross-validation across multiple data splits

LEADERSHIP

Princeton Racket Club

May 2024 - Aug 2024

Tennis Coach and Tournament Director

- Directed tournament logistics; coached athletes across all skill levels in a cross-functional environment; managed client communications and conflict resolution end-to-end

Tennis Racket Stringing Services

Jan 2019 - Present

- Founded and scaled to **50+ loyal clients**; manage all operations, inventory, and client consultations